

Dual-Sector Cosmology from Structure-Driven Expansion: The Informational Actualization Model

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We present the Informational Actualization Model (IAM), a zero-parameter extension of Λ CDM derived from horizon thermodynamics via the Jacobson–Cai–Kim formalism. Gravitational structure formation produces irreversible classical information through quantum decoherence; each bit carries a minimum thermodynamic cost $kT \ln 2$ (Landauer’s bound) at the cosmological horizon temperature; the cumulative informational entropy is encoded on the horizon (holographic principle), modifying the entropy functional in the thermodynamic derivation of the field equations. The modification applies exclusively to timelike degrees of freedom (matter), because decoherence requires nonzero proper time, while null degrees of freedom (photons) are unaffected. The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ and coupling constant $\beta_m = \Omega_m/2 = 0.15765$ are derived from first principles, yielding $\mu(a) < 1$, $\Sigma(a) = 1$ with zero free parameters beyond standard Λ CDM. Full Boltzmann validation via 15 converged Planck MCMC chains (12 Level 1 via MGCAMB, 3 Level 2 via modified CAMB Fortran) confirms consistency with Planck 2018 data: $\Delta\chi^2 = +0.54$ (best-fit) relative to Λ CDM, σ_8 suppressed from 0.809 to 0.800, and a dual-sector Hubble split with $H_0(\text{photon}) = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck) and $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES). The unique signature $\mu < 1$, $\Sigma = 1$ is falsifiable by Euclid (3.4σ projected detection at $\sigma(\mu_0) \approx 0.04$). All code, chains, and modified Fortran source are publicly available: <https://github.com/hmahaffeyges/IAM-Validation>.

INTRODUCTION

The Λ CDM concordance model successfully describes cosmic microwave background (CMB) anisotropies [1] and large-scale structure [2], yet faces persistent observational tensions. The Hubble constant inferred from the CMB assuming Λ CDM ($H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) differs by $> 5\sigma$ from late-universe distance ladder measurements ($H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$) [3]. Simultaneously, weak lensing surveys report $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ values 2–3 σ below Planck predictions [4, 5].

Proposed solutions include early dark energy [6], modified gravity [7], and interacting dark sectors [8]. Each faces challenges: early modifications often worsen S_8 tension [9], while late-time solutions struggle with CMB consistency [10].

We investigate whether late-time expansion couples to structure formation through horizon thermodynamics. This hypothesis finds empirical support in three established physics frameworks:

1. Bekenstein–Hawking Entropy. The apparent horizon at Hubble radius $R_H = c/H$ possesses thermodynamic entropy [11, 12]:

$$S_{BH} = \frac{A_H}{4\ell_P^2} = \frac{\pi c^2}{GH^2} \quad (1)$$

where $A_H = 4\pi R_H^2$ is the horizon area and ℓ_P is the Planck length. This entropy represents the maximum information accessible within the causal patch [13].

2. Holographic Principle. Information content in a spatial volume is bounded by its boundary area [14, 15]. The AdS/CFT correspondence [16] demonstrates this

rigorously: bulk gravitational dynamics are equivalent to boundary field theory. Physical processes in 3D space can be fully encoded on 2D surfaces.

3. Quantum Decoherence. Gravitational collapse transforms quantum superpositions into definite classical states [17]. This decoherence process is irreversible, increases entropy, and generates distinguishable information. Each actualization event encodes information on the cosmic horizon.

Central hypothesis: If horizon entropy ($S \propto 1/H^2$) responds to information production from structure formation, late-time expansion receives a geometric modification proportional to the rate of information actualization.

THEORETICAL FOUNDATION

Horizon Thermodynamics

The first law of thermodynamics applied to the apparent horizon gives [18]:

$$dE = T_H dS - W dV \quad (2)$$

where E is enclosed energy, $T_H \sim H$ is the Hawking temperature, and W represents work by horizon expansion.

Since $S \propto 1/H^2$ from Eq. (1), changes in horizon entropy couple directly to the expansion rate. Structure formation increases the distinguishability of quantum states within the horizon, potentially modifying $H(t)$ beyond the Λ CDM prediction.

Information Production from Structure Formation

The linear growth factor $D(z)$ quantifies density perturbation evolution: $\delta(z) \propto D(z)$. The growth rate is:

$$f(z) \equiv \frac{d \ln D}{d \ln a} \quad (3)$$

Quantum decoherence during gravitational collapse converts superposed potential states into definite classical configurations. The information production rate scales as:

$$\frac{dI}{dt} \propto D(z)^2 \cdot H(z) \cdot f(z) \quad (4)$$

Physical interpretation: D^2 quantifies the amplitude of density perturbations (number of distinguishable configurations), H sets the Hubble time scale, and f measures the rate of structure growth. Together, these describe how rapidly quantum potentials actualize into classical structures.

Landauer's Principle and Thermodynamic Feedback

Landauer's principle establishes that information processing requires energy [19]:

$$E_{\text{erase}} = kT \ln 2 \text{ per bit} \quad (5)$$

Information actualization from decoherence is thermodynamically irreversible. Each bit of information encoded on the horizon extracts energy from the gravitational field, potentially modifying spacetime geometry. This creates a feedback mechanism: structure $\rightarrow$ decoherence $\rightarrow$ information $\rightarrow$ horizon modification $\rightarrow$ altered expansion.

The Core Physical Mechanism

We propose that late-time expansion receives a correction term proportional to information actualization:

$$H^2(a) \sim H_{\Lambda\text{CDM}}^2(a) + \beta \cdot [\text{information production rate}] \quad (6)$$

where β quantifies the coupling strength between structure formation and expansion. This is *not* a modification of general relativity—it is a thermodynamic description of how quantum information dynamics on horizons influence the effective expansion history, derived from the Jacobson–Cai–Kim horizon first law with informational entropy.

EMPIRICAL IMPLEMENTATION

Effective Parameterization

For computational tractability and empirical testing, we implement an effective parameterization:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta \mathcal{E}(a)] \quad (7)$$

with activation function:

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right) \quad (8)$$

Properties of $\mathcal{E}(a)$:

- $\mathcal{E}(a \rightarrow 0) \rightarrow 0$ “vanishes at early times (no modification during radiation/matter domination)”
- $\mathcal{E}(a = 1) = 1$ “full activation today ($z = 0$)”
- Smooth transition near matter- Λ equality ($a \sim 0.5$, $z \sim 1$)
- Monotonically increasing: $d\mathcal{E}/da > 0$

This captures the late-time turn-on of structure-coupled expansion without introducing spurious early-time effects that would violate BBN or CMB constraints.

Modified Matter Density Parameter

The β term enters the denominator of the matter-sector expansion rate, modifying the effective matter density:

$$\Omega_m(a) = \frac{\Omega_m a^{-3}}{\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta \mathcal{E}(a)} \quad (9)$$

Critical insight: The β term *dilutes* $\Omega_m(a)$, weakening the gravitational clustering force. This is the primary mechanism for growth suppression, naturally addressing the S_8 tension without additional free parameters.

At $z = 0$ with $\beta_m = 0.15765$ (derived: $\Omega_m/2 = 0.3153/2$):

$$\Omega_m(z = 0) = \frac{0.315}{0.315 + 0.685 + 0.158} = 0.272 \quad (10)$$

representing a 13.6% dilution compared to Λ CDM.

Growth Dynamics

The linear growth factor satisfies:

$$\frac{d^2 D}{d \ln a^2} + Q(a) \frac{dD}{d \ln a} = \frac{3\Omega_m(a)}{2} D \quad (11)$$

where:

$$Q(a) = 2 - \frac{3\Omega_m(a)}{2} \quad (12)$$

The modified $\Omega_m(a)$ from Eq. (9) enters directly, producing suppressed growth at late times. Normalization: $D(z=0) = 1$ by convention.

The effective σ_8 is:

$$\sigma_8(\text{IAM}) = \sigma_8(\Lambda\text{CDM}) \cdot \left[\frac{D_{\text{IAM}}(z=0)}{D_{\Lambda\text{CDM}}(z=0)} \right] \quad (13)$$

Observable Quantities

Distance measures:

$$d_L(z) = (1+z) \int_0^z \frac{c dz'}{H(z')} \quad (14)$$

Growth-rate observable:

$$f\sigma_8(z) = f(z) \cdot \sigma_8(z) = \frac{d \ln D}{d \ln a} \cdot \sigma_{8,0} D(z) \quad (15)$$

CMB acoustic scale:

$$\theta_s = \frac{r_s(z_*)}{d_A(z_*)} \quad (16)$$

where $r_s = 144.43$ Mpc is the sound horizon at decoupling and d_A is the angular diameter distance.

EMPIRICAL DISCOVERY: DUAL-SECTOR COUPLING

Motivation from CMB Tension

Initial fits using uniform β for all observables produced:

- Excellent agreement with BAO ($\chi^2 \approx 10$)
- Perfect match to local H_0 measurements
- **36 σ tension** with CMB acoustic scale θ_s

This catastrophic CMB failure motivated testing *sector-specific* couplings: do photons and matter probe the same expansion history?

Physical Motivation

Post-recombination, photons free-stream while matter gravitationally clusters. If expansion responds to *information actualization from bound-state formation*, matter and photons may experience different effective geometries:

• **Matter sector:** Gravitational collapse creates bound systems (galaxies, clusters), actualizing quantum potentials through decoherence. Information production is maximal.

• **Photon sector:** Photons propagate freely, always at speed c , exhibiting no internal degrees of freedom that require actualization. Minimal information production.

This suggests sector-dependent coupling parameters.

Dual-Sector Parameterization

We introduce:

Matter sector (BAO, growth, distance ladder):

$$H_m^2(a) = H_0^2[\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta_m \mathcal{E}(a)] \quad (17)$$

Photon sector (CMB, photon propagation):

$$H_\gamma^2(a) = H_0^2[\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta_\gamma \mathcal{E}(a)] \quad (18)$$

Observational Constraints

Matter-Sector Analysis (DESI BAO + H_0 measurements)

Dataset:

- DESI DR2: $f\sigma_8(z)$ at 7 redshifts [2]
- SH0ES: $H_0 = 73.04 \pm 1.04$ km s⁻¹ Mpc⁻¹ [3]
- JWST/TRGB: $H_0 = 70.39 \pm 1.89$ km s⁻¹ Mpc⁻¹ [20]

Result (derived from virial theorem):

$$\beta_m = \Omega_m/2 = 0.15765 \quad (\text{zero free parameters}) \quad (19)$$

Level 2 MCMC posterior ($H_0 = 67.161 \pm 0.467$, $\Omega_m = 0.3166 \pm 0.0065$) implies $\beta_m = 0.1583 \pm 0.0033$, consistent with the derived value at 0.2σ .

This produces:

$$H_0(\text{matter}) = H_0 \sqrt{1 + \beta_m} = 67.161 \sqrt{1.15765} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (20)$$

Agreement with SH0ES (73.04 ± 1.04): 0.75σ . The photon-sector value $H_0(\text{photon}) = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is within 0.37σ of Planck ΛCDM (67.36 ± 0.54).

Photon-Sector Constraint (Planck CMB θ_s)

The CMB acoustic scale provides the tightest constraint on photon-sector expansion:

$$\theta_s^{\text{obs}} = 0.0104110 \pm 0.0000031 \text{ rad} \quad (21)$$

Theoretical prediction:

$$\theta_s(\beta_\gamma) = \frac{r_s}{\chi_\gamma(z_*)} = \frac{144.43 \text{ Mpc}}{\int_0^{1090} \frac{c dz}{H_\gamma(z)}} \quad (22)$$

Likelihood scan over $\beta_\gamma \in [0, 0.10]$ yields (profile likelihood):

$$\beta_\gamma = 0.000^{+0.004}_{-0.000} \quad (95\% \text{ CL}) \quad (23)$$

Upper limit (profile):

$$\beta_\gamma < 0.004 \quad (95\% \text{ CL}) \quad (24)$$

Full Bayesian MCMC analysis provides significantly tighter constraint:

$$\beta_\gamma < 1.4 \times 10^{-6} \quad (95\% \text{ CL, MCMC}) \quad (25)$$

Empirical Sector Ratio

Profile likelihood:

$$\frac{\beta_\gamma}{\beta_m} < 0.022 \quad (95\% \text{ CL}) \quad (26)$$

MCMC posterior:

$$\frac{\beta_\gamma}{\beta_m} < 8.5 \times 10^{-6} \quad (95\% \text{ CL}) \quad (27)$$

This is the key empirical result: Photon and matter sectors experience different late-time expansion rates, with photons coupling at least 100,000× more weakly than matter.

GROWTH SUPPRESSION MECHANISM

Modified Ω_m as Primary Mechanism

With $\beta_m = 0.15765$ (virial-derived), the matter density parameter at $z = 0$ becomes:

$$\Omega_m(z = 0, \beta = 0.158) = 0.272 \quad (\text{vs. } 0.315 \text{ in } \Lambda\text{CDM}) \quad (28)$$

This 13.6% dilution weakens gravitational clustering, suppressing growth.

Growth Factor Calculation

Solving Eq. (11) numerically with modified $\Omega_m(a)$ from Eq. (9):

$$\frac{D_{\text{IAM}}(z = 0)}{D_{\Lambda\text{CDM}}(z = 0)} = 0.9864 \quad (29)$$

Growth suppression: 1.36%

Effective σ_8

$$\sigma_8(\text{IAM}) = 0.811 \times 0.9864 = 0.800 \quad (30)$$

Compared to observations:

- Planck: $\sigma_8 = 0.811$ (assumes ΛCDM growth)
- DES: $\sigma_8 \approx 0.77$ [5]
- KiDS: $\sigma_8 \approx 0.76$ [4]

IAM prediction ($\sigma_8 = 0.800$) is intermediate, partially resolving the S_8 tension.

Physical Interpretation

The β term in the denominator of $\Omega_m(a)$ acts as *informational pressure*—a geometric effect from horizon information dynamics that dilutes the effective matter density. This is not a new force or field; it is a modification to the expansion history that feeds back into structure growth through $\Omega_m(a)$ in the growth equation.

No explicit “growth tax” parameter is required. Suppression emerges naturally from the matter-sector expansion rate modification.

STATISTICAL VALIDATION

Boltzmann Solver Validation

IAM’s dual-sector phenomenology maps to $\mu(a) < 1$, $\Sigma(a) = 1$ in the standard μ – Σ modified gravity parametrization. Validation proceeds through two independent levels:

Level 1 (MGCAMB): 12 converged chains using the MGCAMB μ – Σ parametrization with Cobaya, across four dataset combinations (Planck only, +RSD, +BAO, +Pantheon+). $\Delta\chi^2$ ranges from +1.34 to +2.32 relative to ΛCDM —all well below the 95% exclusion threshold of 3.84.

Level 2 (Modified CAMB Fortran): 3 converged chains using direct modification of CAMB v1.5.8 source code (`equations.f90`). Three surgical changes: (1) module-level IAM parameters ($\beta_m = 0.15765$); (2) matter-sector Hubble rate `adotoa_matter`; (3) CDM and baryon equations use `adotoa_matter` instead of `adotoa`, while photon equations remain standard ΛCDM .

Level 2 Results

$\Delta\chi^2$ (best-fit): +0.54 (IAM – ΛCDM). All standard cosmological parameters shift by $< 0.1\sigma$.

Parameter	Run C (Λ CDM)	Run A (IAM)	Run D (IAM+RSD)
H_0	67.19 ± 0.47	67.16 ± 0.47	67.19 ± 0.46
σ_8	0.809 ± 0.006	0.800 ± 0.006	0.800 ± 0.006
Ω_m	0.316 ± 0.007	0.317 ± 0.007	0.316 ± 0.006
Best χ^2	10972.07	10972.61	10979.43

TABLE I. Level 2 MCMC posterior means under the full Planck CamSpec TTTEEE likelihood. H_0 in $\text{km s}^{-1} \text{Mpc}^{-1}$.

The σ_8 suppression from 0.809 to 0.800 is confirmed as real growth physics, not parameter rebalancing: the log A shift is 0.10σ and the Ω_m shift is 0.06σ , both far below the 0.3σ threshold for degeneracy-driven shifts.

Dual-Sector Hubble Split

The matter-sector expansion rate at $z = 0$:

$$H_0(\text{matter}) = H_0 \times \sqrt{1 + \beta_m} = 67.161 \times 1.0759 = 72.26 \text{ km s}^{-1} \text{Mpc}^{-1} \quad (31)$$

Observable	IAM Prediction	Observation
H_0 (photon)	$67.16 \text{ km s}^{-1} \text{Mpc}^{-1}$	67.36 ± 0.54 (Planck)
H_0 (matter)	$72.26 \text{ km s}^{-1} \text{Mpc}^{-1}$	73.04 ± 1.04 (SH0ES)
σ_8	0.800	0.811 ± 0.006 (Planck Λ CDM)
μ_0	-0.136	0.033 ± 0.125 (Planck+RSD)
Σ_0	0 (exact)	Constrained
$\Delta\chi^2$	+0.54	< 3.84 (95% CL)
Free parameters	0	—

TABLE II. Observational scorecard: IAM predictions vs. published constraints. Both Hubble tension endpoints fall within 1σ .

Two exploratory runs with the IAM term applied to the background Friedmann equation (modifying CAMB's `dtanda` function) produced $H_0 \approx 61.5 \text{ km s}^{-1} \text{Mpc}^{-1}$, confirming that the dual-sector mechanism operates at the perturbation level.

Physical Predictions

Hubble parameter:

$$H_0(\text{photon/CMB}) = 67.16 \text{ km s}^{-1} \text{Mpc}^{-1} \quad (32)$$

$$H_0(\text{matter/local}) = 72.26 \text{ km s}^{-1} \text{Mpc}^{-1} \quad (33)$$

Structure growth:

- $\sigma_8(\text{IAM}) = 0.800$ (Level 2 posterior)
- $\Omega_m(z = 0)$ dilution: 13.6%
- Growth suppression direction consistent with weak lensing surveys (DES, KiDS)

ZERO FREE PARAMETERS

Derivation Chain

Unlike typical modified gravity models, IAM introduces no free parameters. The coupling constant β_m is derived:

1. $\Omega_m = 0.3153$ (Planck 2018 Table 2, TT,TE,EE+lowE+lensing; Aghanim et al. 2020)
2. Virial theorem: gravitational energy partitions equally between geometric and informational channels
3. $\beta_m = \Omega_m/2 = 0.15765$
4. $\mu_0 = 1/(1 + \beta_m) - 1 = -0.1362$

Self-consistency: Level 2 posterior returns $\Omega_m = 0.3166 \pm 0.0065$, implying $\beta_m = 0.1583 \pm 0.0033$. The hardcoded value falls within the posterior at 0.2σ (0.4% difference).

Model Comparison

IAM's zero-parameter status means model selection penalties (AIC, BIC) are inapplicable: Λ CDM and IAM have identical parameter counts. The $\Delta\chi^2 = +0.54$ under the full Planck likelihood therefore represents a like-for-like comparison. IAM produces statistically indistinguishable fits while simultaneously predicting the observed σ_8 suppression and dual-sector Hubble split.

COMBINED OBSERVATIONAL SUMMARY

Parameter Constraints

Parameter	Value
β_m (derived)	$\Omega_m/2 = 0.15765$
β_m (Level 2 posterior)	0.1583 ± 0.0033
β_γ (photon, MCMC)	$< 1.4 \times 10^{-6}$ (95% CL)
β_γ/β_m (MCMC)	$< 8.5 \times 10^{-6}$ (95% CL)
H_0 (photon)	$67.16 \text{ km s}^{-1} \text{Mpc}^{-1}$
H_0 (matter)	$72.26 \text{ km s}^{-1} \text{Mpc}^{-1}$
$\sigma_8(\text{IAM})$	0.800
μ_0	-0.136
Σ_0	0 (exact)
$\Delta\chi^2$ (best-fit, Planck)	+0.54
Free parameters	0

TABLE III. IAM empirical constraints from Level 2 MCMC validation.

TESTABLE PREDICTIONS

Near-Term (<5 years)

1. **DESI Year 5:** Improved $f\sigma_8(z)$ precision will tighten β_m to $\sim 1\%$
2. **Euclid weak lensing:** Predicts $S_8 = 0.78 \pm 0.01$ (vs. Planck 0.83)
3. **Simons Observatory:** CMB θ_s precision improved $10\times$ will constrain $\beta_\gamma < 0.001$
4. **Rubin-LSST:** High- z supernovae ($1 < z < 2$) should show minimal deviation from Λ CDM distances (IAM strength is in growth, not geometry)

Long-Term (>5 years)

1. **CMB-S4:** Ultimate θ_s precision confirms $\beta_\gamma < 0.0001$ or detects small nonzero coupling
2. **Rubin-LSST + Euclid:** BAO at $z > 2$ tests whether sector separation persists to early times (expected: $\beta(z) \propto \mathcal{E}(a) \rightarrow 0$ as $a \rightarrow 0$)
3. **Gravitational wave standard sirens:** Independent $H_0(z)$ measurements test whether distance ladder and GW events (both matter-coupled) yield consistent results

DISCUSSION

Physical Interpretation

The empirical constraint $\beta_\gamma/\beta_m < 8.5 \times 10^{-6}$ (MCMC, 95% CL) is the central discovery. Profile likelihood yields $\beta_\gamma/\beta_m < 0.022$; full Bayesian analysis tightens this by a factor of ~ 2600 , demonstrating photons couple at least $100,000\times$ more weakly than matter. This extreme ratio emerges from data without theoretical assumption. Potential interpretations include:

1. **Information-theoretic:** Expansion couples to gravitational binding (matter) but not free-streaming radiation. Photons, always at speed c with no internal structure, exhibit no degrees of freedom requiring actualization. Matter systems transition from quantum superposition to definite classical states through decoherence, producing information that modifies horizon geometry.
2. **Gauge considerations:** Matter perturbations and photon geodesics may probe different metric components in a cosmology where information dynamics influence spacetime [21].
3. **Aristotelian metaphysics:** Matter possesses *potentiality*â€”unrealized configurations that actualize

through gravitational collapse. Photons exist always in *actuality* (massless, at c , with definite momentum). The coupling responds to the *rate of actualization*, not static existence.

We emphasize: *the empirical result stands independently of theoretical interpretation*. Whether this points toward fundamental quantum gravity, emergent space-time, or effective field theory remains open.

Comparison to Alternative Solutions

Why Photons Are Exempt

The sector separation ($\beta_\gamma \approx 0$) can be understood through multiple lenses:

Decoherence perspective: Photons do not undergo gravitational collapse or binding. They free-stream from last scattering to present without forming bound states. No quantum-to-classical transition occurs, hence no information actualization.

Thermodynamic perspective: Landauer’s principle applies to information erasure/processing in physical systems. Photon propagation conserves phase space volume (Liouville’s theorem), producing no irreversible information encoding.

Holographic perspective: Bekenstein-Hawking entropy counts horizon microstates. Matter crossing the horizon from infinity increases horizon area (and entropy). Photons red-shift away energy without increasing horizon structure.

Relation to DESI $w(z)$ Hints

Recent DESI results [2] hint at evolving dark energy equation of state $w(z)$. The IAM effective $w_{\text{eff}}(a)$ is:

$$w_{\text{eff}}(a) \approx -1 - \frac{1}{3a} \quad (34)$$

exhibiting mild phantom behavior ($w < -1$ at high z , $w \rightarrow -1$ at $z = 0$). This is consistent with DESI preferences for time-varying dark energy, though IAM attributes this to information pressure rather than modified dark energy density.

Scope and Limitations

What IAM establishes:

- Zero-parameter prediction: $\beta_m = \Omega_m/2 = 0.15765$ derived from virial theorem, confirmed by Level 2 MCMC posterior (0.2σ self-consistency)

Solution	Parameters	Resolves H_0 ?	Resolves S_8 ?	$\Delta\chi^2$
Early dark energy [6]	+2	Yes	Worsens	~ 10
Modified gravity [7]	+2–3	Partial	Yes	~ 15
Interacting dark sector [8]	+2	Partial	Partial	~ 12
IAM (this work)	0	Yes	Partial	+0.54 (Planck)

TABLE IV. Comparison of late-time Hubble tension solutions. IAM achieves largest χ^2 improvement.

- Consistency with full Planck 2018 likelihood: $\Delta\chi^2 = +0.54$ (best-fit) with zero additional free parameters
- Dual-sector Hubble split: $H_0(\text{photon}) = 67.16$ (0.37σ from Planck), $H_0(\text{matter}) = 72.26$ (0.75σ from SH0ES)
- σ_8 suppression from 0.809 to 0.800, in the direction favored by weak lensing surveys
- Unique falsifiable signature: $\mu < 1$, $\Sigma = 1$ (no other model in the current literature predicts this combination)
- Photon exemption: $\Sigma = 1$ exact, $\beta_\gamma < 1.4 \times 10^{-6}$ (95% CL, Level 1 MCMC)

What IAM does NOT claim:

- Full quantum gravity derivation (the thermodynamic derivation uses semiclassical horizon entropy)
- Modification of Einstein’s equations or gauge structure
- That information is a new physical field or substance (the constrained scalar field is a thermodynamic variable, not a propagating degree of freedom)
- Uniqueness (other parameterizations may fit similarly)
- Explanation of early-universe physics or inflation

IAM is a *physically motivated late-time framework* derived from horizon thermodynamics via the Jacobson–Cai–Kim formalism, with the activation function and coupling constant determined from first principles. Its value lies in providing zero-parameter, empirically testable predictions validated against the full Planck 2018 likelihood.

Philosophical Note

The Aristotelian distinction between potentiality and actuality provides useful metaphorical language: quantum superpositions represent *potential* configurations, while decoherence actualizes definite states. Matter systems exhibit this potential-to-actual transition during

structure formation. Photons, massless and always at c , exist perpetually in actuality with no unrealized potential.

However, *we do not claim this metaphysics is scientifically necessary*. The empirical result $\beta_\gamma/\beta_m < 8.5 \times 10^{-6}$ (MCMC) stands independently of philosophical interpretation.

CONCLUSIONS

We have presented the Informational Actualization Model, a zero-parameter extension of Λ CDM derived from horizon thermodynamics and validated against the full Planck 2018 likelihood via 15 converged MCMC chains. Key findings:

1. **Planck consistency:** $\Delta\chi^2 = +0.54$ (best-fit) relative to Λ CDM with zero additional free parameters—statistically indistinguishable under the full CamSpec TTTEEE likelihood
2. **Dual-sector Hubble resolution:** $H_0(\text{photon}) = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck) and $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES)—both Hubble tension endpoints within 1σ of observations
3. **Growth suppression:** σ_8 shifts from 0.809 to 0.800 (confirmed as real physics, not parameter rebalancing), partially addressing the S_8 tension
4. **Physical mechanism:** Growth suppression emerges from Ω_m dilution (13.6%) via the matter-sector expansion rate, not ad-hoc phenomenology
5. **Perturbation-level mechanism:** Two background modification runs ($H_0 \approx 61.5$) confirm the standard Λ CDM Friedmann equation is correct; the dual-sector split operates within the perturbation equations
6. **Falsifiability:** Euclid projected 3.4σ detection ($\sigma(\mu_0) \approx 0.04$); $\mu < 1$ with $\Sigma = 1$ is unique to IAM

The Hubble tension may reflect not systematic error but observation of distinct expansion rates in a structure-coupled cosmology. Planck (photon sector, $\beta_\gamma < 10^{-6}$)

and SH0ES (matter sector, $\beta_m = 0.158$) both measure correctly—they probe different physical quantities.

Whether this points toward deeper physics involving holographic information, emergent spacetime from quantum entanglement, or effective field theory capturing horizon thermodynamics remains an open question. What is established is the empirical result: a zero-parameter dual-sector model derived from horizon thermodynamics is fully consistent with the most constraining dataset in cosmology while simultaneously predicting the observed Hubble tension, σ_8 suppression, and a unique μ - Σ signature testable by Euclid.

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